

Machine Learning Project Report: Loan Approval Prediction

For: Alfido Tech

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1. Project Objective

To build a Machine Learning model that predicts whether a customer's loan will be approved (1) or rejected (0) based on their demographic and financial profile. The goal is to automate the decision-making process while minimizing the financial risk of defaults.

2. Data Preprocessing & Feature Engineering

- **Handling Missing Values:** Missing categorical data was filled using the mode, and continuous data using the median to ensure robustness against outliers.
- **Categorical Encoding:** Converted text data (Gender, Married, Education, etc.) into machine-readable numeric formats using `pd.get_dummies()` (One-Hot Encoding), dropping the first column to prevent multicollinearity.
- **Train-Test Split:** Divided the data into 80% training and 20% testing sets using stratified sampling to maintain the target distribution.

3. Scaling & Handling Class Imbalance

- **Selective Scaling:** Applied `StandardScaler` strictly on continuous numerical columns (like Income and LoanAmount) to prevent distortion of binary categorical columns (0s and 1s).
- **Preventing Data Leakage:** Scaling was applied *after* the train-test split (`fit_transform` on Train, only `transform` on Test) to ensure the model does not get any hints about the unseen test data.

- **SMOTE (Synthetic Minority Over-sampling):** The dataset had a heavy imbalance (more Approvals than Rejections). SMOTE was applied *only* on the scaled training data to synthetically balance both classes and prevent the model from becoming biased towards approvals.

4. Model Evaluation & Comparison

Two models were trained on the balanced data and evaluated on the unseen Test data:

Metric	Logistic Regression	Random Forest
Precision (Class 1)	0.84	0.84
Recall (Class 1)	0.84	0.78
F1-Score (Class 1)	0.84	0.81
ROC-AUC	0.83	0.80
Overall Accuracy	77%	74%

Conclusion: Logistic Regression outperformed the more complex Random Forest model. It successfully maintained a stronger balance between Precision and Recall.

5. Business Interpretation & Risk Analysis

In the banking sector, errors have different costs:

- **False Positives (High Risk):** Predicting a loan as "Approved" when the customer will actually default. This causes direct financial loss to the principal amount.
- **False Negatives (Low Risk):** Predicting a loan as "Rejected" when the customer was capable of paying. This results in a missed interest-earning opportunity.
- **Current Performance:** The winning model successfully identifies 61% of potential defaulters (Class 0 Recall), but 39% are still slipping through as false positives.

6. Final Recommendation & Deployment Strategy

By default, Logistic Regression uses a 0.5 (50%) probability threshold to approve a loan.

- **Actionable Insight:** For a conservative lending strategy, the bank should increase the decision threshold from 0.50 to 0.60 or 0.65.
- **Impact:** The model will only approve a loan if it is more than 60-65% confident. This will significantly reduce the number of False Positives (bad loans approved) and protect the bank's capital, which is the ultimate priority in credit risk management.